Introduction

2 minutes

In a short few years, generative AI, a subset of artificial intelligence that focuses on creating new content, has changed the way we work and revolutionized what is possible with technology. At times, the fast-moving developments in generative AI can feel challenging to keep track of even for seasoned developers.

In this module, gain a framework for understanding generative AI applications and how Microsoft Foundry supports innovation. What does today's innovation look like? Consider these use cases:

- **Marketing Content Creation**: Companies use Microsoft Copilot's generative AI to automatically write product descriptions, blog posts, and social media content—saving time and ensuring brand consistency across platforms.
- **Customer Support**: Businesses deploy AI-powered virtual agents that can understand and respond to customer inquiries in natural language, offering 24/7 support and reducing the load on human agents.
- **Code Generation**: Developers use tools like GitHub Copilot to generate code snippets, suggest functions, and even write entire modules based on natural language prompts, speeding up software development.
- **Image and Video Generation**: Designers and content creators use the latest models in Microsoft Foundry's model catalog t generate visuals for campaigns, storyboards, or concept art—often from just a text description.

- **Personalized Learning and Tutoring**: Educational platforms use generative AI to create custom quizzes, explanations, and study guides tailored to a student's learning style and progress.

Microsoft offers an ecosystem of tools for AI use and development. This module explores the Foundry model catalog and how to discover, evaluate, and deploy an appropriate model. Learn how to test and configure the deployed model the Foundry playground, and call it from code using the OpenAI-compatible Responses API. Finally, you'll see how agents encapsulate a model, its instructions, and optional tools so your solution is reusable and consistent across Playground and code via the Project API.

> **Note**
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> We recognize that different people like to learn in different ways. You can choose to complete this module in video-based format or you can read the content as text and images. The text contains greater detail than the videos, so in some cases you might want to refer to it as supplemental material to the video presentation.

Next unit: Generative AI models

Next